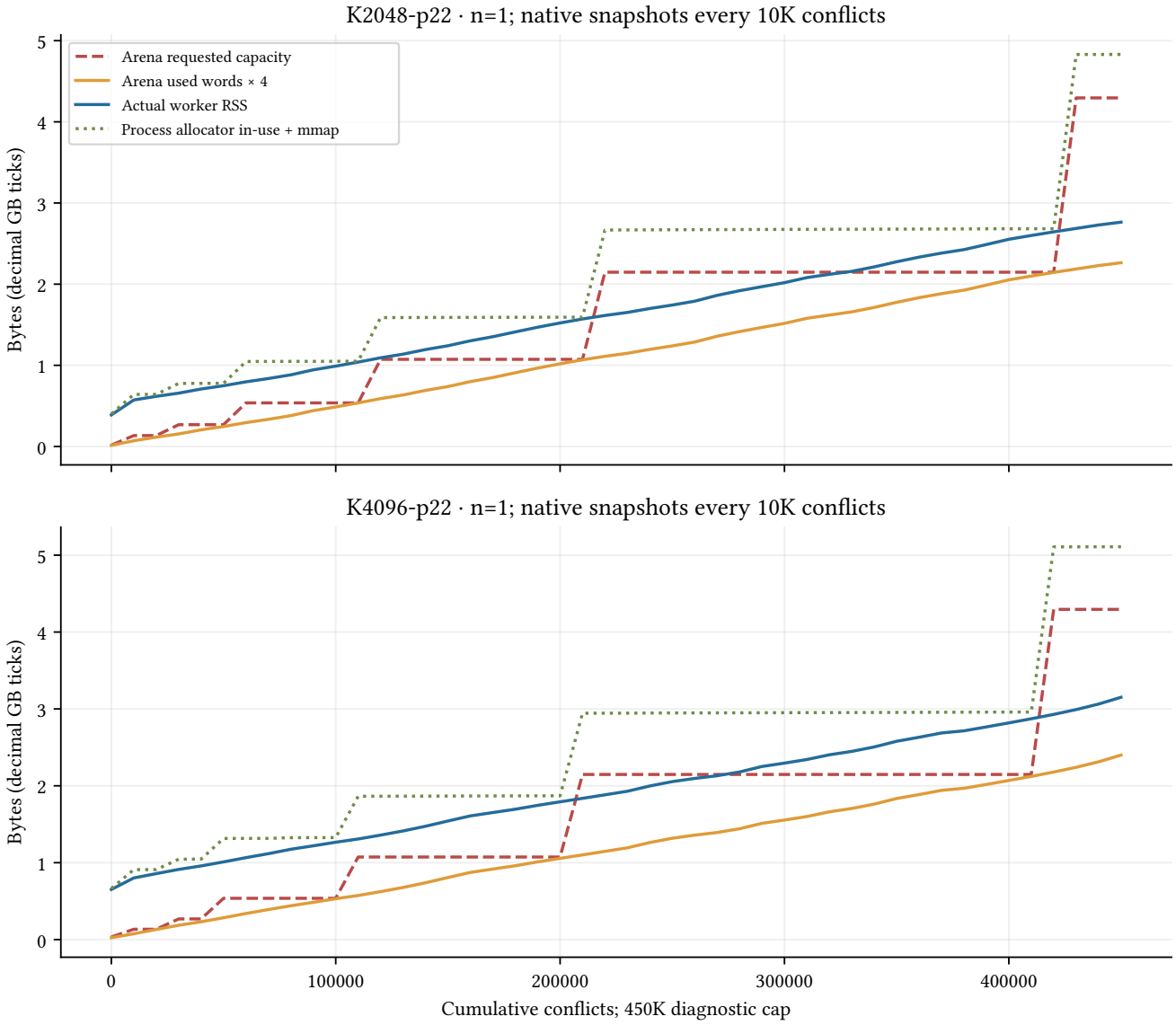
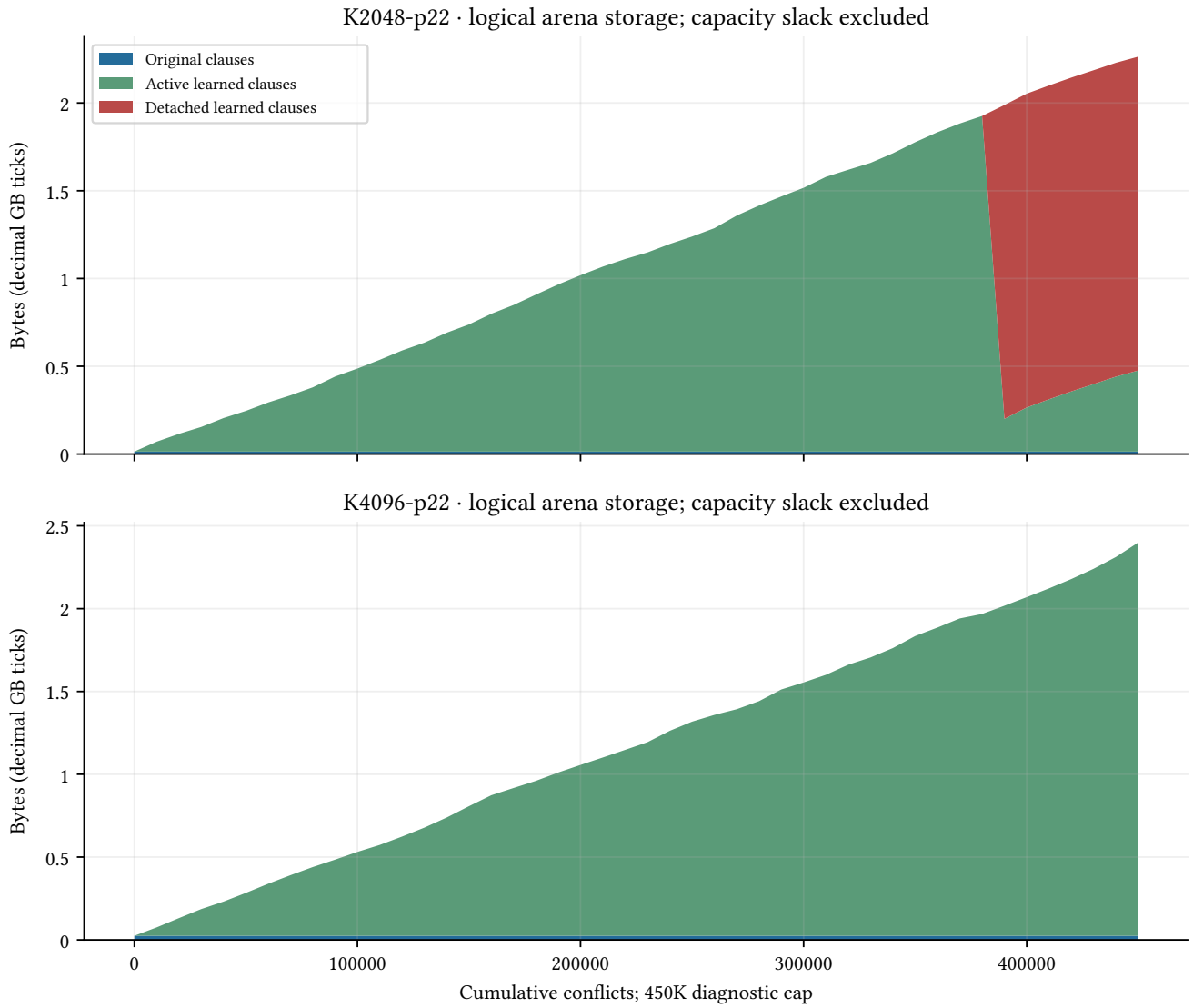
Native N83 SAT storage ownership

Diagnostic campaign2026-10-21UTC. Public retained K2048/K4096 panels, 450000conflicts/2400worker-seconds each,8GiB/zero-extra-swap. A separate allocation diagnostic after both original1M arms completed; it does not replace or extend that sweep's search cap. n=1 per base, no uncertainty estimate. No instrumented clock is admitted as a matched timing comparison.



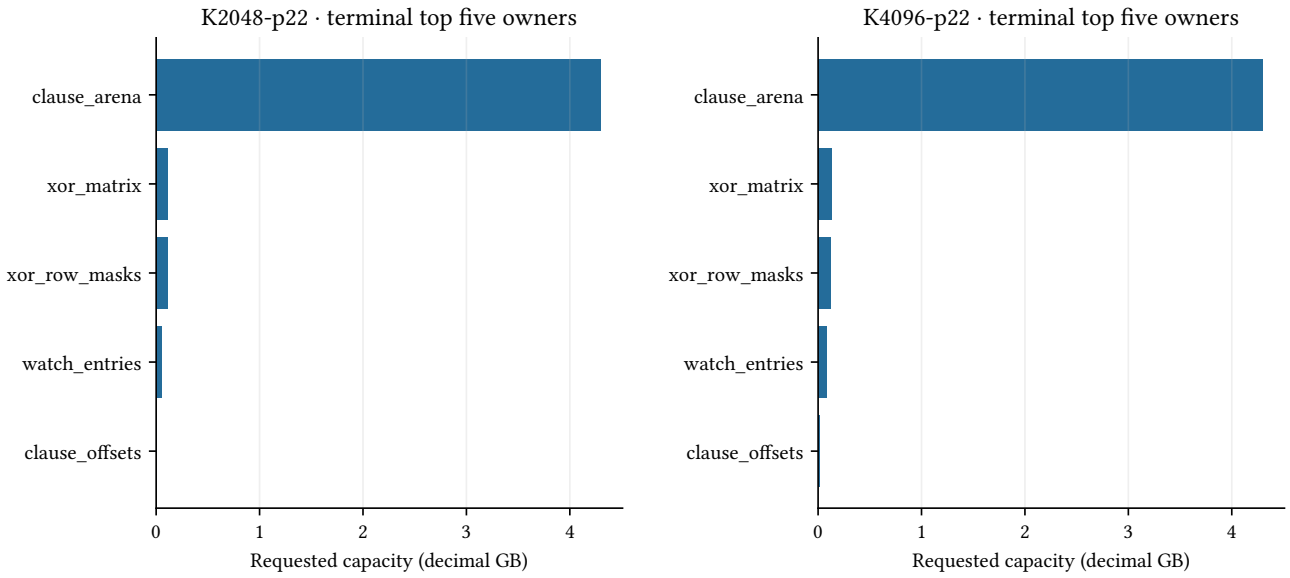
Requested Vec buffer capacity, logical arena words and actual SAT-worker RSS are distinct. Process glibc in-use-plus-mmap includes non-solver storage and uncommitted capacity; no per-buffer page attribution or allocation-stack trace. Snapshot schedule: solve entry/return and each10000conflicts. Native checker validates all24 components, sums, arena partitions, worker PID/RSS/HWM units, full trace boundaries and retained conflict-counter signatures.

Retained learned-clause storage



The appendix/native JSON records exact byte counts. K2048 crosses its first clause reduction; K4096 remains before its first retained reduction threshold. An absence of detached K4096 clauses at450K says nothing about larger caps. Requested capacities are not resident bytes; allocator retention is not itself a leak. No reclamation candidate has run in this diagnostic campaign.

Owners, execution and limits



Frozen worker SHA256: c4b2062df1bd14e13c2c3d9adae28bdb552be103a0c6dd3f2dbc272ce3b1337d. Worker source674d2075958c5cb388cbfcb271e8ebcd6dea646c adds read-only telemetry; SAT algorithm/branching/forgetting unchanged. K2048 uses guard-v21; K4096 uses queued guard-v22 after a retained infrastructure refusal. Same worker bytes, public inputs,image,core14/30,workerCPU14,1CPUquota,8GiB/no extra swap, strict quiet/eviction/restoration,networknone,read-only/capdropALL/64PIDs. Host ip-172-31-26-135.us-east-2.compute.internal, Xeon6975P-C,32logicalCPU. Missing per-worker context-switch/fault counters and fixed kernel threads prevent automatic L2. No task build/test/compression ran during its workers.

K2048-p21 shell redirection failed before native execution: missing parent, no receipt/worker/duration; retained director impeded assessment. K2048-p22 completed450K/zero. K4096-p21 command exited1 after10.985894322seconds, nonwaiting lock refused before setup; all14 outputs missing. Guard-v22 adds explicit600-second bounded interruptible lock queue before CPU planning, with hashed policy/elapsed record, same lock/permissions and fixed worker cap. Original refusals remain unadmitted. Exact relays, receipts, raw SAT traces, cgroup samples,startup records and cleanup are retained outside this report.

Evidence and next step

Study research/koblitz_n83_factor_base_sweep_20261008; proof verification/native-allocation-profile-20261021. REPORT.md and PUBLICATION.md link actual per-arm values, source/ELF/relay identities, native summaries, all failures/tests and S3 PUT/HEAD/freshGET/payload/replay/repack proofs. Graphs render native JSON from data/ with same-source150dpiPNG. Report PDF visually inspected before publication; benchmark copies have identical bytes. Original1M and older failed attempts retain their separately verified archives.

Canonical docs/ic-leaderboard.html, docs/index-calculus-scoreboard.html and docs/ic/progress-timeline.json remain unchanged: no new eligible cold/rho/ identity/component timing promotion. Full cold accounting, natural relations, rank/recovery and broader PR2036 CI/readiness remain open. Existing CMS5.17.0 known-witness external comparison remains faster; no fastest-tool claim or new matched natural external comparison. Next: separately preregister and test arena reclamation preserving clause/reason references and retained search signatures, then measure before extending caps. This report grants no speedup, UNSAT, recovery or minimum-memory claim.

Exact terminal observations

Bytes unless stated	K2048-p22	K4096-p22
Requested solver Vec capacity	4611260673	4673328420
Arena requested capacity	4294967296	4294967296
Arena capacity / all owned	0.9314084803637385	0.9190381907719638
Arena logical used	2263844356	2399593924
Original clauses	14108924	24717564
Active learned clauses	461943344	2374876360
Detached learned clauses	1787792088	0
Worker RSS at return	2763960320	3151409152
Worker HWM at return	2764464128	3151409152
Process allocator in-use+mmap	4830762864	5109477536
XOR reason literals	2136400	2108032
Matched conflict snapshots	451	451
Storage snapshots	47	47
Cgroup samples	29	29
Greatest sampled cgroup charge	2746028032	3022114816
Last sample to outer end(ms)	27008.30878400011	38622.77860700013
Command wall(seconds)	1721.016573635	1731.917490259

Native capacities and fractions are derived from exact recorded owner sums and arena partitions; no Python research computation. Conflict snapshots include zero. Both native terminals450000conflicts/models0/relations0/rank0,exit0/noOOM/ no wall censor. All sampled8GiB/swap0; sampled kernel peaks are lower bounds on full-run peaks. Worker RSS may exceed the last cgroup sample. Original1M runs remain complete in their own immutable evidence; these450K diagnostics do not replace or extend them.

Actual mechanical relay SHA256: K2048 c2b64ea540c876bd5336eafee62272e5555a16f6d41a2191d83b4e540b6e0ca2 K4096 6d32f93a890c5a4aad92214fd3da123840438bba66f3d0a34694b3e6586c202f Director assessments grant execution compliance; root independently verifies full audits, profiles, storage invariants and all retained conflict signatures. Both read-only live Docker captures equal final raw logs byte-for-byte. Amended conservative native-card envelope7200worker-seconds includes full2400 for K4096′s refused command; actual two-worker query allowance4800seconds. Earlier shell failure has missing measured duration, not fabricated zero. This remains separate from incomplete historical campaign accounting.

Full release validation at code source7e8ae96c:4471library pass/0fail/119ignored; 19exporter pass;2independent model-checker pass;38native guard unit+7card pass. Initial telemetry-specific readiness31SAT tests and old guard35+7 remain retained. Python is used only for plotting/rendering/PDF inspection. Full native positive/negative replay from freshly downloaded final S3 bytes is established by external PUBLICATION.md receipts, not an assertion inside this PDF.